

Precipitation

Top End Thunderstorms

Required condition	Why essential
<i>i.</i> Atmospheric cooling to dew-point (by convective uplift in Darwin)	Lowers saturation vapour pressure; allows condensation to start.
<i>ii.</i> Abundant condensation nuclei (sea-salt, smoke from bushfires)	Provide surfaces for initial microscopic droplet formation; without them, air can remain supersaturated.
<i>iii.</i> Droplet growth to $>\approx 3mm$ (collision–coalescence in warm clouds)	Gives droplets sufficient mass to overcome up-draughts and fall as rain.

Table 1

If any one is missing, vapour remains suspended, clouds persist without rainfall, or virga evaporates before reaching ground.

Singapore vs Scottish Highlands

- Singapore $\Rightarrow$ **collision–coalescence** in warm ($> 0\text{ }^{\circ}\text{C}$) cumulonimbus; Highlands $\Rightarrow$ **Bergeron (ice-crystal) process** in mixed-phase nimbostratus.
- Singapore: whole cloud above freezing, strong up-draughts $\rightarrow$ droplets of different radii collide & merge. Highlands: vertical cross-section spans $-5\text{ }^{\circ}\text{C}$ to $-15\text{ }^{\circ}\text{C}$; ice crystals grow at expense of super-cooled droplets because $e_{s,ice} < e_{s,liq}$; moderate lifting in frontal system.
- Warm-cloud rain gives **large, uniform raindrops**, often $> 2mm$, causing high-intensity bursts ($> 100\text{ mm h}^{-1}$). Bergeron process yields **smaller frozen aggregates**, which melt on descent $\rightarrow$ lower intensity but longer-lasting precipitation, with mixed graupel or wet snow.

Sierra Nevada

- Orographic precipitation:
 - Max orographic band = 1 000–3 000 m on windward (western) flank.
 - Rain shadow = Owens Valley east of crest ($< 300 \text{ mm yr}^{-1}$).
- Dry-adiabatic: $9.8 \text{ }^{\circ}\text{C km}^{-1}$ up to LCL ($\approx 1200 \text{ m}$), then saturated rate $6 \text{ }^{\circ}\text{C km}^{-1}$ to 3000 m: $\Delta T \approx -23 \text{ }^{\circ}\text{C}$. Descent: warms at dry rate: $+29 \text{ }^{\circ}\text{C}$ to valley floor.
- Example towns: **Bishop, Lone Pine**. Both import water via Los Angeles Aqueduct; pumping from deep alluvial fans carefully managed because local recharge is minimal under rain-shadow conditions. [See link.](#)

Clearing versus Forest Experiment

```

1  import pandas as pd
2  import matplotlib.pyplot as plt
3
4  df = pd.read_csv('Paired_Gauges_Table.csv', parse_dates=['Date'])
5  df.plot(x='Date', kind='bar', width=0.85)
6  plt.ylabel('Daily rainfall (mm)')
7  plt.title('Forest vs Pasture Gauge - Pará, 10-23 Feb')
8  plt.xticks(rotation=45, ha='right')
9  plt.legend(['Forest', 'Pasture'])
10 plt.tight_layout()
11 plt.show()

```

Total rainfall: Forest = 140 mm; Pasture = 119 mm ($\approx +15\%$).

Possible biases & fixes:

Bias / micro-site factor	Likely effect	Mitigation
Dense canopy causes interception loss before drops reach gauge orifice.	Under-catch at Forest site.	Install <i>trough</i> throughfall collectors + stemflow collars to quantify interception; or raise gauge above understory in a tower.

Wind-turbulence differences: clearing exposed → eddies may enhance Pasture catch.	Over-catch at Pasture.	Equip both gauges with Alter shields, measure local wind speed for correction factors.
Splash-in from drip points off overhanging leaves near Forest gauge.	Over-catch on intense days, but net effect variable.	Clear 1 m radius around gauge; fit non-splash grid.

Table 2

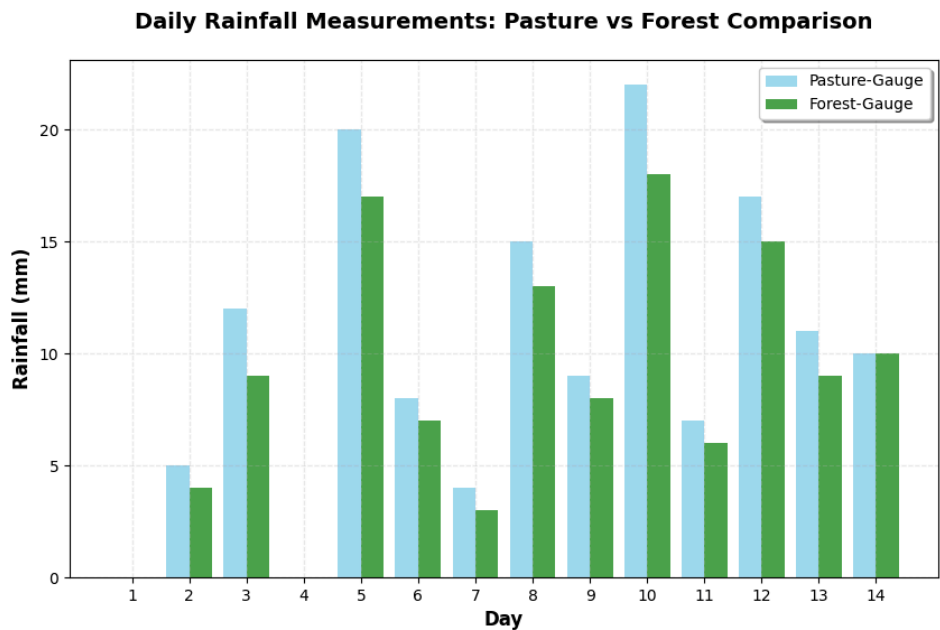


Figure 1: Paired bar chart of rainfall at Forest and Pasture gauges.

A 5 Himalayan Catchment

- **Hypsometric method** chosen:
 - Because gauges cluster low ($< 1\,800\text{ m}$) yet $\approx 40\%$ of basin lies above snowline, rainfall–altitude gradient is strong & non-linear. Thiessen would over-weight lowlands; hypsometric belts (e.g., 500 m intervals) allocate snowfall intensities based on elevation and aspect → better represent orographic input and snow-water storage.
- Supplement **GPM (Global Precipitation Measurement) IMERG** satellite or regional weather-radar composites. Calibrate bias using available gauges; blend with hypsometric zones through kriging or Bias-Corrected Precipitation Reconstruction (BCPR) to refine daily grids.